Economics S5040: Dynamic Stochastic Modeling
Problem Set 6 — The Solow Growth Model and U.S. Calibration

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Instructions

Submit one Jupyter notebook containing all derivations, Python code, figures, numerical answers, and economic explanations.

This problem set has two goals. First, you will work with the Solow growth model analytically. Second, you will turn the model into a quantitative model of the United States using Penn World Table (PWT) 11.0 data.

For the empirical problems, obtain the PWT 11.0 Excel file from the official Penn World Table website and save it as `pwt110.xlsx` in the same directory as your notebook.

Unless otherwise stated, use the U.S. observations from 1990 through 2019 as the benchmark calibration period. We stop the benchmark period in 2019 so that the unusual pandemic years do not determine the calibrated parameters.

A numerical answer without an economic explanation will receive little credit.

Problem 1. Capital Accumulation, the Steady State, and Convergence

Consider the Solow model with constant labor and Cobb–Douglas production

$$y_t = Ak_t^{\alpha}, \qquad 0 < \alpha < 1.$$

A constant fraction s of output is invested, and capital depreciates at rate δ. The law of motion is

$$k_{t+1} = sAk_t^{\alpha} + (1-\delta)k_t.$$

(a) Derive the positive steady-state capital stock k^*.

(b) Derive steady-state output per worker y^* and consumption per worker c^*.

(c) Show that

$$k_{t+1} - k_t = sAk_t^{\alpha} - \delta k_t.$$

Use this equation to explain why $k_t < k^*$ implies $k_{t+1} > k_t$, and why the inequality reverses when $k_t > k^*$.

(d) Let

$$A = 1, \qquad \alpha = 0.33, \qquad s = 0.25, \qquad \delta = 0.08.$$

Compute k^*, y^*, and c^*.

(e) Starting once from $k_0 = 0.25k^*$ and once from $k_0 = 2k^*$, simulate 75 periods. Plot both paths together with the steady state. Explain what the figure demonstrates about convergence.

Problem 2. Population Growth, Comparative Statics, and Factor Prices

Now suppose employment grows at the constant rate n:

$$L_{t+1} = (1 + n)L_t.$$

Aggregate capital still obeys

$$K_{t+1} = (1 - \delta)K_t + sY_t.$$

(a) Show that capital per worker obeys

$$k_{t+1} = \frac{sAk_t^\alpha + (1 - \delta)k_t}{1 + n}.$$

(b) Show that a positive steady state satisfies

$$sA(k^*)^\alpha = (n + \delta)k^*,$$

and therefore

$$\boxed{k^* = \left(\frac{sA}{n + \delta}\right)^{1/(1-\alpha)}}.$$

(c) Holding all other parameters fixed, determine the direction in which k^* changes when each of s, A, δ, n rises. Explain the economics, not just the algebra.

(d) Under competitive Cobb–Douglas production, show that

$$w = (1 - \alpha)Ak^\alpha, \qquad r = \alpha Ak^{\alpha-1} - \delta.$$

If an economy accumulates capital, what happens to w and r? Explain.

(e) In the no-population-growth case, show that the saving rate that maximizes steady-state consumption is $s_{\text{GR}} = \alpha$. Why is this result not yet a theory of household saving?

Problem 3. Reading the Penn World Table

Load PWT 11.0 and keep only observations for the United States. For this assignment use

`rgdpna`, `rnna`, `emp`, `labsh`, `delta`, `rtfpna`.

Here `rgdpna` is a real-GDP series based on national-accounts growth rates, `rnna` is the real capital-stock series at constant national prices, `emp` is employment, `labsh` is labor's share of income, `delta` is PWT's depreciation rate, and `rtfpna` is PWT's national-accounts TFP index.

(a) Display the first five and last five U.S. observations for these variables. Report the first and last years available.

(b) Plot U.S. real GDP, real capital, and employment over time.

(c) Construct

$$y_t = \frac{Y_t}{L_t}, \qquad k_t = \frac{K_t}{L_t}, \qquad \frac{K_t}{Y_t}.$$

Plot all three series.

(d) Using 1990–2019, calibrate

$$\boxed{\hat{\alpha} = 1 - \text{mean}(\texttt{labsh}_t)}.$$

Report $\hat{\alpha}$ and the average labor share.

(e) Calibrate average employment growth using

$$\boxed{\hat{n} = \text{mean}\left(\frac{L_{t+1}}{L_t} - 1\right)}.$$

Report $\hat{n}$.

Problem 4. Using Capital Accumulation to Calibrate s and δ

For aggregate capital,

$$K_{t+1} = (1 - \delta_t)K_t + I_t.$$

(a) Rearrange this equation to construct implied gross investment:

$$\boxed{I_t = K_{t+1} - (1 - \delta_t)K_t}.$$

Use `rnna` for K_t and `delta` for δ_t.

(b) Construct

$$\boxed{s_t = \frac{I_t}{Y_t}}.$$

Plot s_t for the United States. Discuss any unusual observations.

(c) Over 1990–2019, compute

$$\hat{s} = \text{mean}(s_t), \qquad \hat{\delta} = \text{mean}(\delta_t).$$

Report both numbers.

(d) The steady-state condition with employment growth implies

$$\frac{K^*}{Y^*} = \frac{s}{n + \delta}.$$

Compare the average U.S. capital-output ratio over 1990–2019 with

$$\frac{\hat{s}}{\hat{n} + \hat{\delta}}.$$

How closely does the simple accumulation model line up with the U.S. data?

Problem 5. Calibrating Productivity and the U.S. Solow Steady State

Use the calibrated value $\hat{\alpha}$ from Problem 3.

(a) Construct the simple Cobb–Douglas productivity residual

$$\boxed{A_t = \frac{y_t}{k_t^{\hat{\alpha}}}}.$$

Plot A_t from 1990 onward.

(b) Let A_{2019} be the value of your residual in 2019. Holding technology fixed at A_{2019} and using $\hat{s}, \hat{\delta}, \hat{n}, \hat{\alpha}$, compute

$$\boxed{k^*_{2019} = \left(\frac{\hat{s}A_{2019}}{\hat{n} + \hat{\delta}}\right)^{1/(1-\hat{\alpha})}}.$$

Compare k_{2019} with k^*_{2019}.

(c) Starting from observed k_{2019}, simulate 75 periods under the fixed parameters and fixed technology A_{2019}. Plot the path and the steady state.

(d) Solve for the same steady state using Newton's method. Define

$$h(k) = \hat{s}A_{2019}k^{\hat{\alpha}} - (\hat{n} + \hat{\delta})k$$

and use

$$k_{m+1} = k_m - \frac{h(k_m)}{h'(k_m)}.$$

Compare the Newton solution with the analytical solution and report the absolute difference.

(e) Explain why the simulation in part (c) should not be interpreted as a literal forecast of U.S. capital per worker for the next 75 years.

Problem 6. The Solow Residual and PWT's TFP Measure

Your residual from Problem 5 uses the simplest production function in the course:

$$Y_t = A_t K_t^{\hat{\alpha}} L_t^{1-\hat{\alpha}}.$$

PWT's `rtfpna` is a more sophisticated growth-accounting measure.

(a) Normalize your constructed residual so that $\widetilde{A}_{2021} = 1$. Plot $\widetilde{A}_t$ and `rtfpna` on the same graph from 1990 onward.

(b) Compute the correlation between the annual log growth rates of the two series.

(c) Are the two series identical? They should not be. Explain at least two reasons why PWT's `rtfpna` can differ from your simple residual. Your answer should recognize that PWT's growth-accounting procedure uses more information than our one-capital-input, raw-employment Cobb–Douglas model.

(d) Using your simple residual, verify numerically

$$\Delta \log y_t = \Delta \log A_t + \hat{\alpha}\, \Delta \log k_t.$$

Report the maximum absolute discrepancy caused by numerical rounding.

(e) Write a short paragraph answering: What did calibration add to the Solow model that was not present when A, s, δ, α, n were simply arbitrary numbers?

What surprised me?

Write one short paragraph describing one empirical fact, numerical result, or model-data discrepancy from this assignment that surprised you. Explain why.